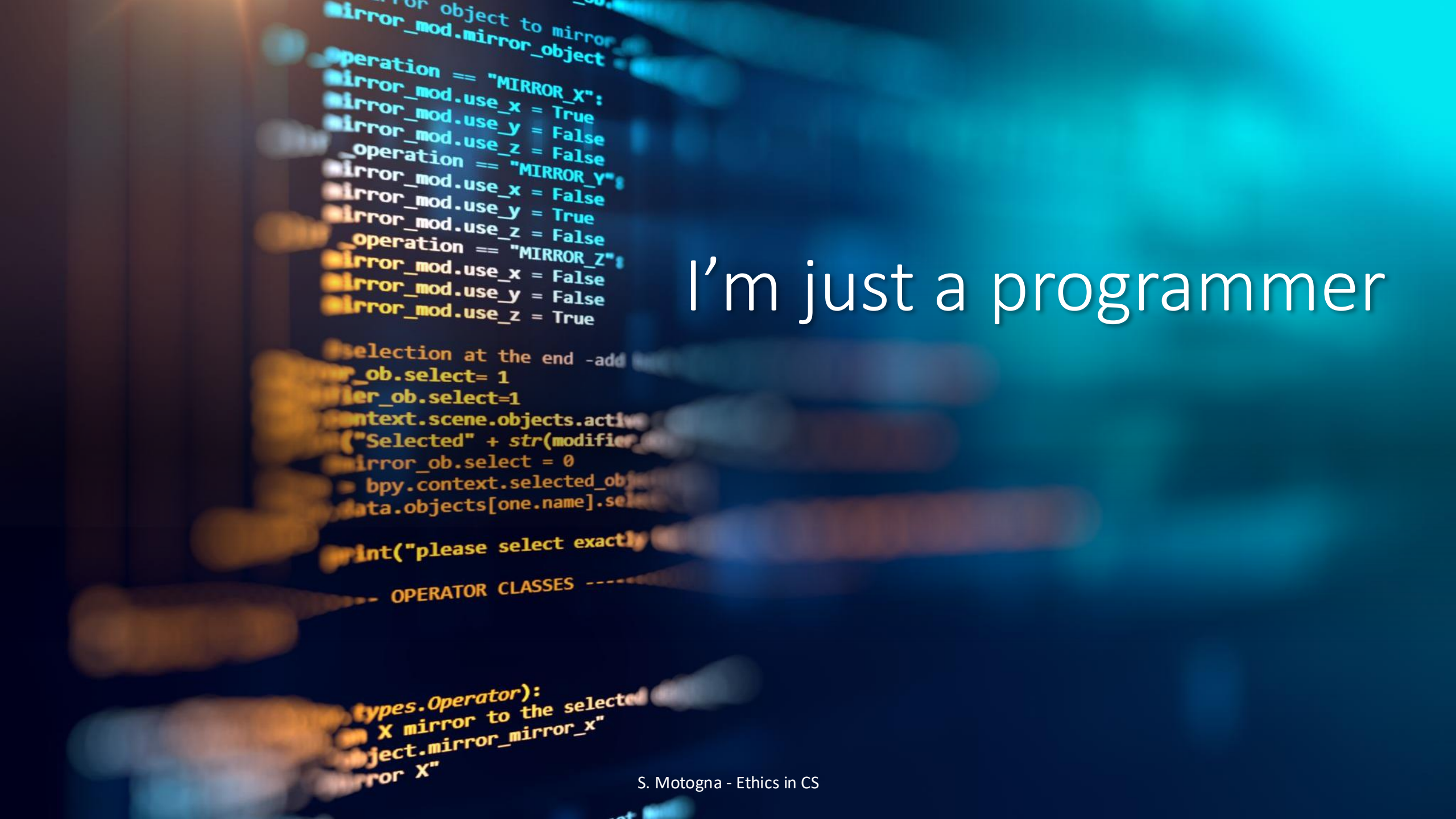
Course 8



Risks and liabilities in software development



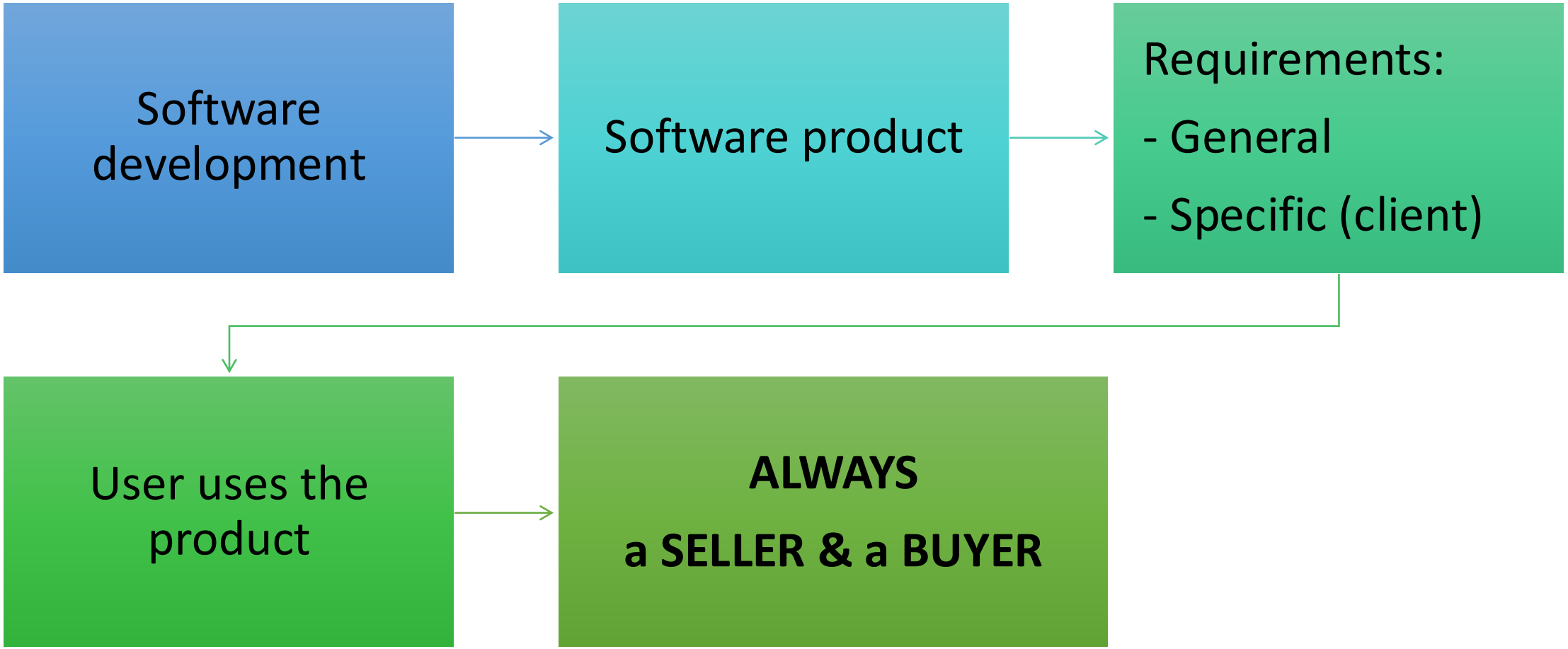
I'm just a programmer

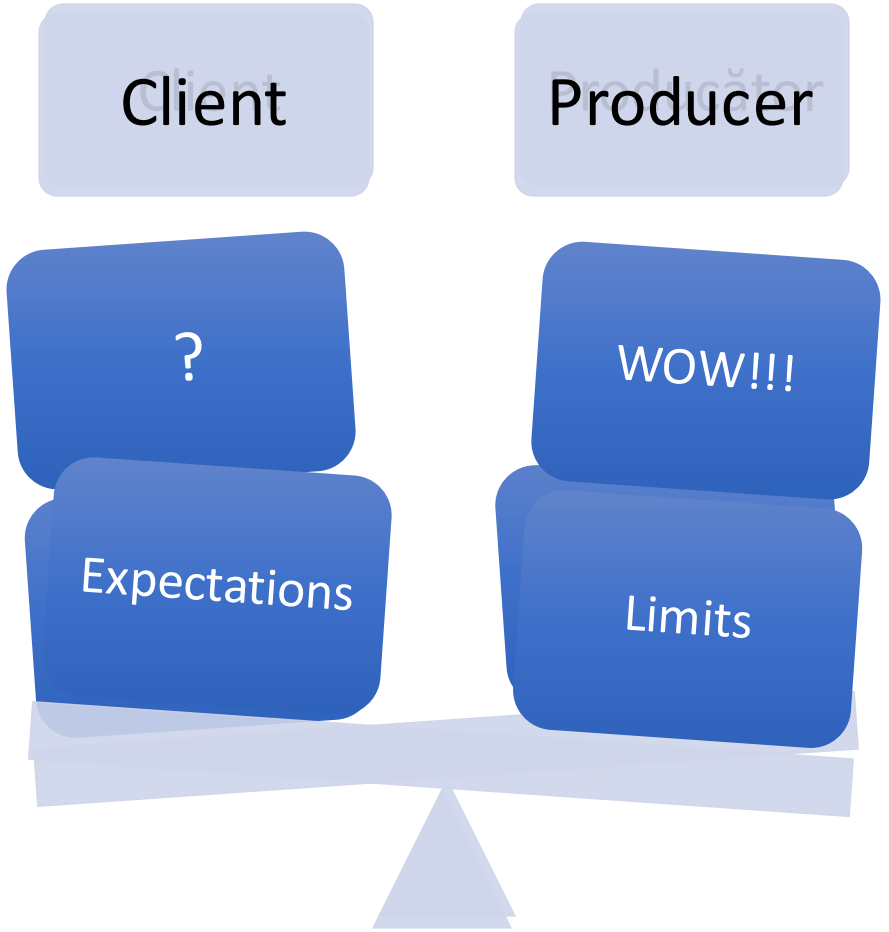
The Ethics of Programming

- Computer scientist Grady Booch writes about programming ethics, and his view is that:



Concepts

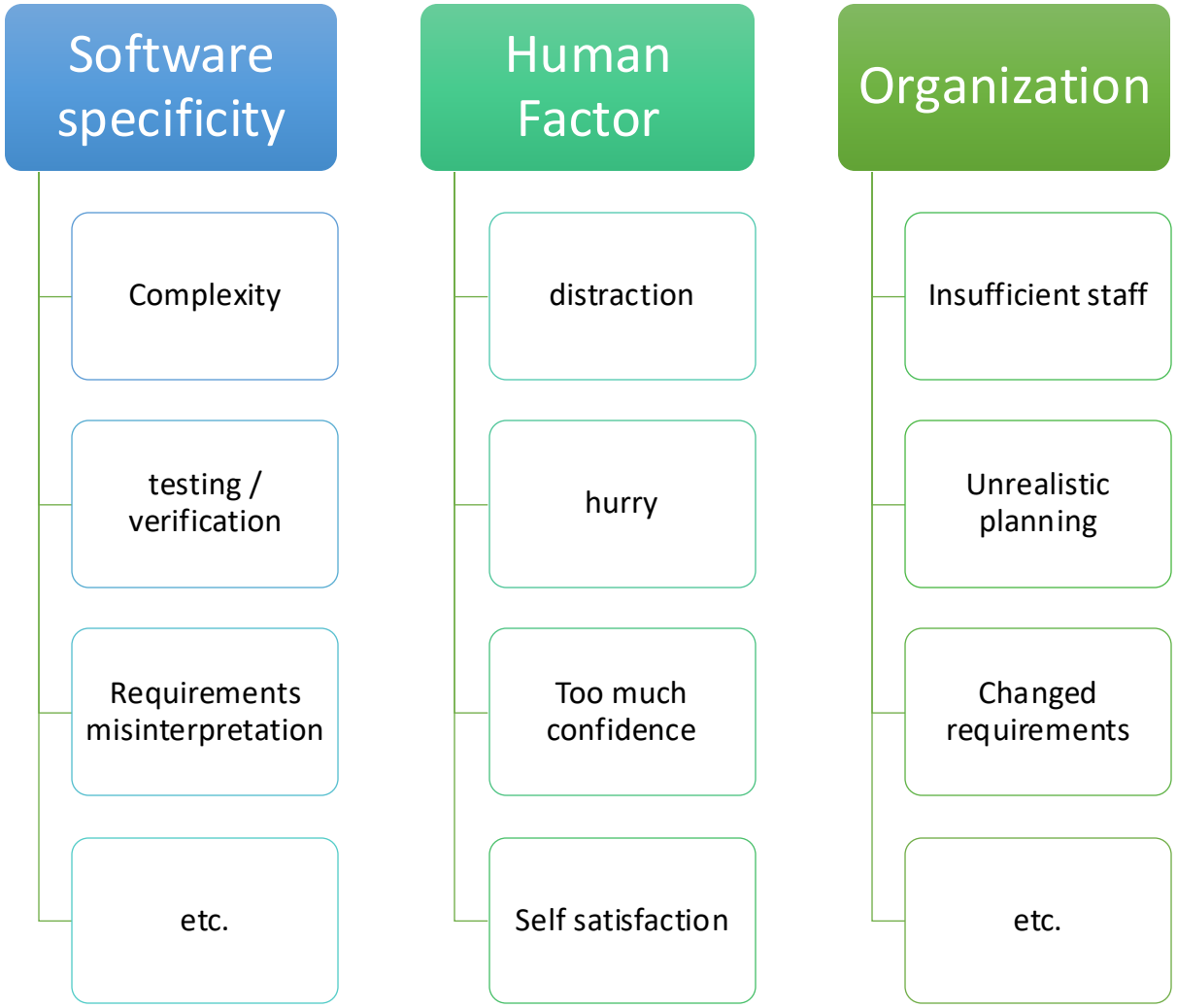

Software development
Software product
Requirements:
- General
- Specific (client)
User uses the product
ALWAYS
a SELLER & a BUYER


Client
Producer
?
WOW!!!
Expectations
Limits

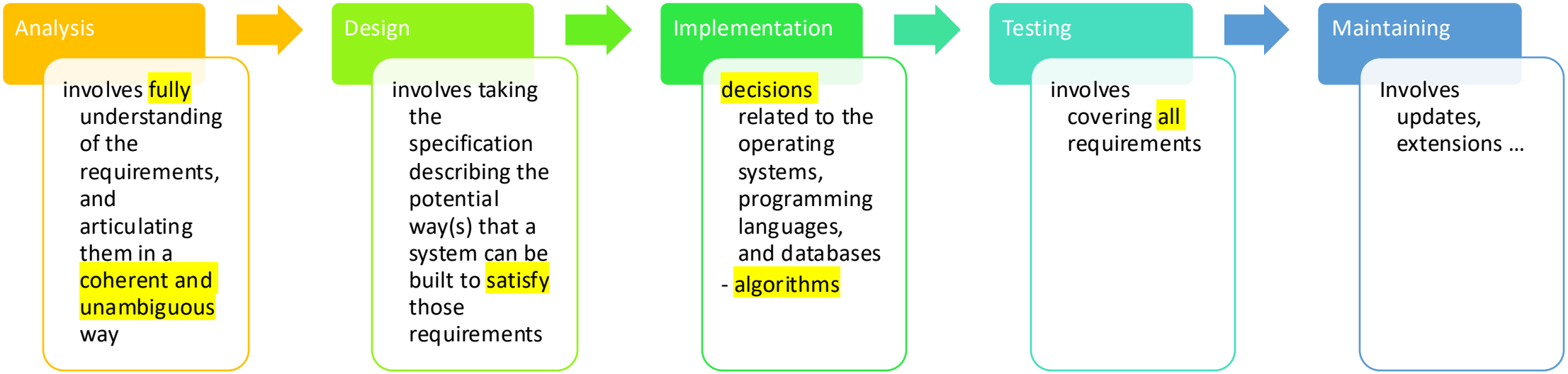
Client – producer relation:



Risks – why?

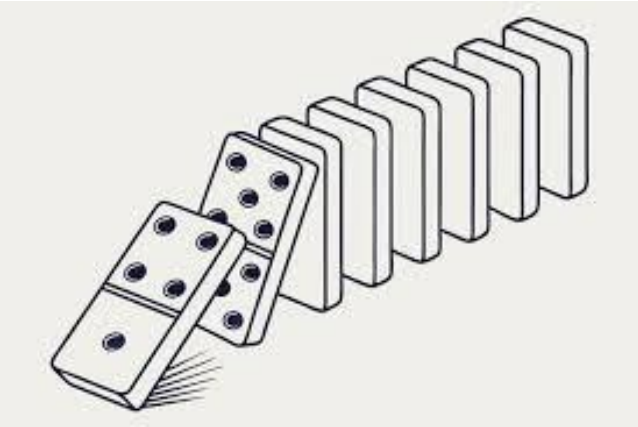

Software specificity
Complexity
testing / verification
Requirements misinterpretation
etc.
Human Factor
distraction
hurry
Too much confidence
Self satisfaction
Organization
Insufficient staff
Unrealistic planning
Changed requirements
etc.

Software development



Domino effect in Software Development

Single unethical or poor-quality decision can trigger a chain reaction of negative consequences, potentially leading to public crises, financial loss, and damage to user trust



Privacy breaches

Security vulnerabilities

Algorithmic bias

System failures

Famous cases

Maintaining

Famous cases: Therac 25

- Computer assisted system for radiation therapy
- After 6 months in use, defects appeared
- 13.000—20.000 rad INSTEAD OF 200 rad
- Deaths
- Cause:
  - Updates to Therac 6 software – software reuse

Design

Implementation

Famous cases: Intel Pentium

- 1994: reports – bug in Intel Pentium micro-processor: some arithmetic operations were incorrect
- Intel denies the problem
- Then, they admit: „it was a problem, the error appears only very complex applications, and most of the programs/users will not even realise it”
- „we are already working on it”
- A large number of buyers request replacing the defect chip
- In the end, public relations are in collapse and Intel agrees to replace

Ethical / legal?

Implementation

Famous cases: Heartbleed Bug

- Security Bug in OpenSSL Criptography library
- Introduced in 2012, publicly recognised in 2014
- Exemple: access patient data

Implementation

Testing

Famous cases: Irish State Exams 2020

- Due to pandemic: no final exams
- Calculator grade: algorithm - estimation using their expected performance combined with the school's statistical profiles of achievement
- Incomplete testing
- Lots of errors

Testing

Implementation

Famous cases: Volkswagen Emission Scandal

- Emission tests running in lab
- Not in real tests

Testing

Maintaining

Famous cases: Crowdstrike & Microsoft 2024

Case 1 – Algorithms

- What are the ethical risks of big data analytics or algorithms, which can sometimes lead to false?

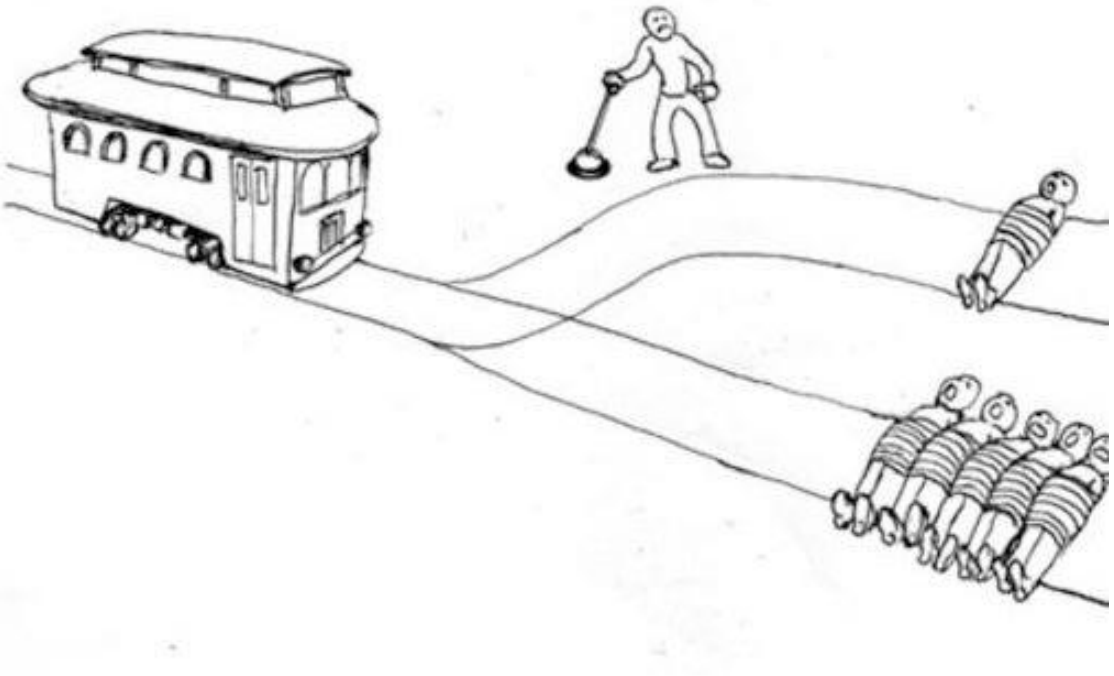
- What might be done to mitigate these risks?

Case 2 – Testing

- “it would be nice to do some more testing, but we just don't have the time. Besides, I think that if anything was wrong, it would have showed up …”

- How much testing?

Autonomous vechiles



https://www.moralmachine.net/